INCH-POUND

MIL-I-24683A(SH)
26 November 1990

SUPERSEDING
MIL-I-24683(SH)
19 May 1987

MILITARY SPECIFICATION

INSULATION ELECTRICAL, HIGH TEMPERATURE INORGANIC PAPER

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a series of flexible electrical insulations for use in applications for electrical machinery at temperatures up to 220 degrees Celsius (°C).

1.2 Classification. Electrical insulation shall be of the following types and forms as specified (see 6.2).

1.2.1 Types. The types of electrical insulation shall consist of the following:

Type I – Inorganic single layer paper with low level of acrylonitrile binder
Type II – Two-ply laminate of type I paper, laminated with an acrylic adhesive
Type III – Inorganic paper with moderate levels of organic binders
Type IV – Same as type III but with higher toughness.

1.2.2 Forms. The forms of electrical insulation paper shall consist of the following:

Form R – Rolls 4 inches or greater in width (see 3.3.2.1)
Form S – Flat sheets (see 3.3.2.2). Rolls 4 inches wide or greater (see 3.3.2.1)
Form T – Rolls less than 4 inches in width (tape) (see 3.3.2.1).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5970

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1.2.3 **Part numbering.** Material shall be identified and ordered by the standard system, M24683-IR05S100. The part numbering system shall be as follows:

M24683-IR05A525

Material quantity per roll or package:
Use three digits – include leading zeros
For forms R and T = average yards per roll
For form S = sheets per package

Material size – as specified in table I (use one letter)

Material thickness:
Use two digits – include leading zero
Digits indicate thickness in thousandth of an inch

Material form – (use one letter):
R = form R, S = form S, T = form T

Material type – (use one digit):
1 = type I, 2 = type II
3 = type III, 4 = type IV

Specification number – (use five digits)

Specification identified (use one letter):
B = military specification for materials in bulk quantities
D = military specification using metric measurement units
M = military specification using US customary units.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 **Specifications and standards.** The following specifications and standards, form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

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SPECIFICATIONS

FEDERAL

MMM-A-260	Adhesive, Water-Resistant, (for Sealing Waterproofed Paper)
PPP-B-1055	Barrier Material, Waterproofed, Flexible
PPP-F-320	Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade) and Cut Shapes
PPP-T-60	Tape: Packaging, Waterproof

MILITARY

MIL-P-116	Preservation, Methods of
MIL-L-10547	Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, flexible
MIL-L-19140	Lumber and plywood, Fire-Retardant Treated

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-130	Identification Marking of US Military Property
MIL-STD-2073-1	DOD Material Procedures for Development and Application of Packaging

(Unless otherwise indicated, copies of federal and military specifications and standards are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

C 177	Standard Test Method for Steady-State Heat Flux Measurements and Thermal Transmission Properties by Means of the Heat Guarded Hot Plate; (DOD adopted)
D 149	Standard Test Method for Dielectric Breakdown Voltage and Dielectric Strength of Solid Electrical Insulating Materials at Commercial Power Frequencies; (DOD adopted)
D 150	Standard Test Methods for A-C Loss Characteristics and Permittivity (Dielectric Constant) of Solid Electrical Insulating Materials; (DOD adopted)
D 202	Standard Methods for Sampling and Testing Untreated Paper Used for Electrical Installation.
D 257	Standard Test Methods for D-C Resistance or Conductance of Insulating Materials; (DOD adopted)
D 374	Standard Test Methods for Thickness of Solid Electrical Insulation
D 618	Standard Methods of Conditioning Plastics and Electrical Insulating Materials for Testing; (DOD adopted)
D 774	Standard Test Method for Bursting Strength of Paper; (DOD adopted)
D 828	Standard Test Method for Tensile Breaking Strength of Paper and Paperboard
D 1830	Standard Test Method for Thermal Endurance of Flexible Sheet Materials Used for Electrical Insulation by the Curved Electrode Method
D 3951	Standard Practice for Commercial Packaging; (DOD adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

TECHNICAL ASSOCIATION OF THE PULP AND PAPER INDUSTRY (TAPPI)

T413 OM-85	Ash in Paper and Paperboard

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(Application for copies should be addressed to the Technical Association of the Pulp and Paper Industry, Technology Park/Atlanta, P0 Box 105113, Atlanta, GA 30348.)

UNDERWRITERS LABORATORIES, INC. (UL)

1446 Systems of Insulating Materials – General

(Application for copies should be addressed to the Underwriters Laboratories, Inc., 333 Pfingsten Road, Northbrook, IL 60062.)

(Non-Government standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.2 Material. The electrical insulation paper shall be asbestos-free and composed of aluminum silicate reinforced with an elastomeric binder. To protect direct current (dc) rotary equipment the insulation shall not contain any organic silicone compounds. This paper shall not contain, or produce when shop processed, any respirable fibers or dust.

3.3 Dimensions.

3.3.1 Thickness. The material shall be in accordance with the thickness requirements shown in table I and shall be provided in the nominal thickness as specified (see 6.2).

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TABLE I. *Thickness and length.*

Nominal thickness (inches)	Minimum thickness (inches)	Maximum thickness (inches)	Minimum length per roll (yards)	Average length per roll (yards)
0.0025	0.0020	0.0030	910	960 ± 50
0.003	0.0025	0.0035	850	800 ± 50
0.0035	0.0030	0.0040	700	725 ± 25
0.0045				
0.005	0.0045	0.0055	500	525 ± 25
0.007	0.0063	0.0077	355	375 ± 19
0.0075				
0.010	0.0090	0.0110	250	265 ± 13
0.015	0.0135	0.0165	165	75 ± 9
0.020	0.0180	0.0220	125	130 ± 5
0.025	0.0225	0.0275	100	105 ± 5
0.030	0.0270	0.0330	83	87 ± 4
0.040	0.0360	0.0440	–[1]	–[1]
0.050	0.0450	0.0550	–[1]	–[1]
0.060	0.0540	0.0660	–[1]	–[1]
0.033	0.0028	0.0038	670	720 ± 50

[1] Since thicknesses 0.040, 0.050, and 0.060 inches are normally furnished as form S materials, no minimum or average length per roll has been established. Type II materials are normally provided in thicknesses 0.040, 0.050, and 0.060 only.

3.3.2 Width and length.

3.3.2.1 Forms R and T. Unless otherwise specified (see 6.2 and 6.4), form R and T materials shall be provided on rolls of the widths (trimmed within the tolerances) specified below. Unless otherwise specified (see 6.2), form R and form T materials, 0.006 through 0.030 inch nominal thickness, shall be provided on 3-inch inside diameter (id) cores, which extend the full width of the material, with full rolls having an 11-1/2 ± 1/4-inch outside diameter (od) and a minimum length of material as indicated in table I. The average length of all form R and T rolls in any lot shall be as indicated in table I. Form R and T materials, 0.040 to 0.060 inches nominal thickness, shall be provided on 6-inch id cores, which extend the full width of the material, with the full roll diameter and the minimum and average length of material per roll as specified (see 6.2). Form R material shall be provided in widths of 4 inches and greater. Form T material shall be supplied in widths of 3/8 to 3-15/16 inches, with each roll having no more than four defects. Defects shall be marked with bright red or yellow pressure-sensitive tape.

Nominal width (inches)	**Width tolerance (inches)**
For forms R and T	
4 through 8	± 1/32
over 8	± 1/16

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3.3.2.2 Form S. Unless otherwise specified (see 6.2), form S material shall be provided in squared and trimmed flat sheets having dimensions of 36 by 36 or 36 by 24 inches, plus or minus 1/8 inch. There shall be no visual defects on the sheets.

3.4 Performance requirements. The material shall be in accordance with the performance requirements specified in tables II and III.

TABLE II. *Performance requirements, thickness related.*

Require-ments	Conditions[1]	Units	Values required										
			Nominal thickness (inches)										
			Types I and II							Type II only			
			0.0035	0.005	0.007	0.010	0.015	0.020	0.025	0.030	0.040	0.050	0.060
Average dielectric strength	168/23/55 Bent lengthwise	Volts per mil (minimum)	150	150	150	150	150	150	100	100	100	100	100
Average dielectric strength	168/23/55 Bent crosswise	Volts per mil (minimum)	150	150	100	100	100	100	100	100	100	100	100
Average tensile strength	168/23/96 Lengthwise	Pounds per inch width (minimum)	3	5	7	10	15	20	24	30	38	45	50
Average tensile strength	168/23/55 Crosswise	Pounds per inch width (minimum)	1	2	3	5	7	8	10	12	15	17	20
Weight loss on ignition	925 °C	Percent (maximum)	32	30	30	30	30	30	30	30	32	32	32
Mullen burst strength	168/23/55	Pounds per square inch (minimum)	3	4	6	9	13	18	22	25	37	45	53

See footnote at end of table.

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TABLE II. *Performance requirements, thickness related – Continued.*

Require-ments	Conditions[1]	Units	Values required									
			Nominal thickness (inches)									
			Types III							Type IV		
			0.0033	0.005	0.0075	0.010	0.0025	0.003	0.0045	0.005	0.0075	0.010
Average dielectric strength	168/23/55 Bent lengthwise	Volts per mil (minimum)	150	150	150	150	150	150	150	150	150	150
Average dielectric strength	168/23/55 Bent crosswise	Volts per mil (minimum)	150	150	150	150	150	150	150	150	150	150
Average tensile strength	168/23/96 Lengthwise	Pounds per inch width (minimum)	10	11	20	25	10	10	14	14	20	25
Average tensile strength	168/23/55 Crosswise	Pounds per inch width (minimum)	4	5	8	10	4	4	5	5	8	10

[1]Conditions – 168/23/55 – 168 hours/23 °C/55 percent relative humidity.

3.5 Identification of sheets for stock. Package of sheets of form S material provided for stock shall be marked with nominal thickness and stock number in accordance with MIL-STD-130.

3.6 Directional marking of sheets. The sheets of insulation shall be marked to show the machine direction.

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TABLE III. *Performance requirements, independent of thickness.*

Requirements	Condition	Units	Values required
Apparent density	168/23/55	Grams/cubic centimeter (maximum)	0.90
Average dielectric strength	168/23/55	Volts per mil (minimum)	150
Average dielectric strength	168/23/96	Volts per mil (minimum)	150
Average dielectric strength	At 220 °C straight	Volts per mil (minimum)	150
Volume resistivity	168/23/96	Ohm/centimeter (minimum avg.)	1×10^{10}
Dissipation factor (60 hertz (Hz))	168/23/96 At 220 °C	(maximum) (maximum)	0.30 0.30
Dielectric constant (60 Hz)	168/23/96 At 220 °C	(maximum) (maximum)	5.0 5.0
Thermal endurance, 10 mil only	As specified in 4.6.8	Temperature °C for 40,000 hours (minimum)	220 (Types I & II) 180 (Types III & IV)
Apparent thermal conductivity	220 °C 180 °C	Btu – in/hr ft^2 °F (minimum)	1.20 (Types I & II) 1.00 (Types III & IV)

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of the manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3)
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. First article inspection shall consist of the examinations and tests specified in table IV. Inspection of form R material of a particular thickness shall constitute qualification of forms S and T material of the same thickness fabricated under essentially the same conditions in the same plant.

4.3.1 Samples for first article inspection. First article samples for each type and thickness shall consist of 10 square yards of form S material.

4.4 Quality conformance inspection.

4.4.1 Visual and dimensional examinations. Each roll or package shall be examined during cutting operation to verify conformance to this specification. Examination for defects shall be visual or by automatic, optical, mechanical, or electrical detector. A defect is a hole or an area that is non-uniform in shade or texture. Also, a defect is an area where impregnation or a coating has occurred with another material. Dimensional examination shall be as specified in 4.6.2. Any roll or package which contains one or more unmarked visual or dimensional defects shall be rejected.

4.4.2 Quality conformance tests. The sample mill roll shall be subjected to the quality conformance tests listed in table IV to verify conformance to this specification. Thickness measurements shall be included in quality conformance testing and shall conform to the requirements of 3.3.1.

4.4.3 Lot. Material of the same thickness, manufactured under essentially the same conditions, and offered for delivery at the same time, shall be considered a lot for purposes of sampling and inspection. The lot size shall be stated in terms of mill rolls. A mill roll is considered a full width machine run roll. Each lot size shall consist of no more than eight mill rolls.

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TABLE IV. *First article and quality conformance inspection.*

Test	Requirement	Test	First article inspection minimum number of determinations per sample unit	Quality conformance inspection minimum determinations per sample unit
Visual examination	3.3	4.4.1	As specified in 4.4.1	As specified in 4.4.1
Dimensions	3.3	4.6.1 and 4.6.2	As specified in 4.6.1 and 4.6.2	As specified in 4.6.1 and 4.6.2
Apparent density	3.4	4.6.3	3	3
Dielectric strength				
Straight – 168/23/55	3.4	4.6.4	5	–
Straight – 168/23/96	3.4	4.6.4	5	5
Straight at 220 °C	3.4	4.6.4	5	–
Bent lengthwise – 168/23/55	3.4	4.6.4	5	–
Bent crosswise – 168/23/55	3.4	4.6.4	5	5
Volume resistivity – 168/23/96	3.4	4.6.5	3	–
Dissipation factor:				
168/23/96	3.4	4.6.6	3	–
At 220 °C	3.4	4.6.6	3	3[1]
Dielectric constant:				
168/23/96	3.4	4.6.6	3	–
At 220 °C	3.4	4.6.6	3	3[1]
Tensile strength:				
Lengthwise – 168/23/55	3.4	4.6.7	5	5
Crosswise – 168/23/55	3.4	4.6.7	5	5
Mullen burst – 168/23/55	3.4	4.6.10	5	–
Thermal endurance	3.4	4.6.8	As specified in 4.6.8	–
Elevated temperature exposure	3.4	4.6.9	–	5
Loss on ignition 1/925	3.4	4.6.11	1	1
Apparent thermal conductivity – 220 °C	3.4	4.6.12	–	–[2]

[1]Dissipation factor and dielectric constant shall be measured at 220 °C for types I and II and 180 °C for types III and IV once every year on samples of each type and thickness of material obtained from normal production runs.
[2]Effective thermal conductivity shall be determined only upon special request.

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4.4.4 Sampling for visual and dimensional examinations. Each roll or package shall be examined as specified in 4.4.1. A roll shall be considered to be a slit portion of a mill roll in widths as specified in 3.3.2.1. A package shall be considered to be a stack of flat sheets as specified in 3.3.2.2.

4.4.5 Sampling for quality conformance tests. Sample mill rolls from each lot produced shall be selected in accordance with MIL-STD-105, general inspection level II, and then subjected to the quality conformance tests specified in 4.4.2.

4.5 Conditioning of test specimens.

4.5.1 Nomenclature. The designations indicating conditioning of test specimens shall be in accordance with ASTM D 618.

4.5.2 Temperature and humidity tolerances. Specified temperatures shall be maintained within the tolerances shown in table V. Specimens shall be tested with moisture contents equal to those obtained in equilibrium with the specified relative humidity, plus or minus the tolerances shown in table V.

TABLE V. *Temperature and humidity tolerances.*

Temperature		Relative humidity	
Range (°C)	Tolerance (plus or minus °C)	Humidity (percent)	Tolerance (plus or minus percent)
23 to 180	2	55	5
181 to 300	3	96	3
301 to 325	4		

4.6 Tests.

4.6.1 Thickness. Specimen thickness shall be measured using a manual or motorized deadweight dial-type micrometer, or its equal, in accordance with method C of ASTM D 374. The pressure exerted on the specimen shall be 23 to 27 pounds per square inch (lb/in^2), and the anvil surface upon which the specimen rests shall be 2 inches in diameter for the manual test and 1/4-inch in diameter for the motorized tests. The thickness shall be determined as the average of 10 readings spaced at intervals along the length or across the width of the sample representing the mill roll or rolls being evaluated.

4.6.2 Width and length. The length and width dimensions of rolls and sheets shall be controlled and audited during forming and cutting operations to provide specified sizes within allowable tolerances, and to document length of material on each mill roll (see 3.3).

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4.6.3 Apparent density. Volume and weight of specimens shall be determined after conditioning 168/23/55. Basis weight shall be determined in accordance with ASTM D 202. Thickness shall be measured in accordance with 4.6.1. The apparent density for each specimen shall be calculated by dividing its weight by its volume.

4.6.4 Dielectric breakdown.

4.6.4.1 Electrical apparatus. Step up transformer, circuit breaker, voltage control, and voltmeter shall be in accordance with ASTM D 149. Power supply frequency shall be between 50 and 100 Hz and the transformer shall have a rating of not less than 2 kilovoltampere (kVA). The short-time-test shall be at a rate-of-rise as follows:

Thickness (inches)	Volts per second
0.002 to 0.010	100
0.015 to 0.025	250
0.030 to 0.060	500

4.6.4.2 Test electrode assemblies.

4.6.4.2.1 Straight specimens. The test electrodes shall consist of a metal cylinder 2 inches in diameter and not less than 1/2 inch in length, with edges rounded to a radius of 1/4 inch. To prevent flash-over, the sample shall extend well away from the outer circumference of the electrodes; or, 1/8-inch thick annular rubber (or silicone rubber) gaskets with 2-1/8 inch diameter center holes shall be employed around the electrodes. The total weight of the upper electrode on the specimen shall be approximately 1 pound.

4.6.4.2.2 Bent specimens. Brass electrodes, 90 degree V-shaped, as shown on figure 1 shall be used. Edges of the electrodes shall be rounded to approximately 1/64-inch radius.

4.6.4.3 Bending jig. A 90 degree, V-shaped jig, as shown on figure 2 shall be used.

4.6.4.4 Method.

4.6.4.4.1 Straight. Specimen size shall be 5 by 5 inches. Specimens shall be conditioned 168/23/55 and 168/23/96. Dielectric breakdown shall then be measured on the specimens at room conditions of 23 °C within 10 minutes after removal from the humidity conditioning chamber. The specimens shall be maintained at 220 °C for 8 ± 3 minutes, and then shall be subjected to dielectric breakdown at this temperature.

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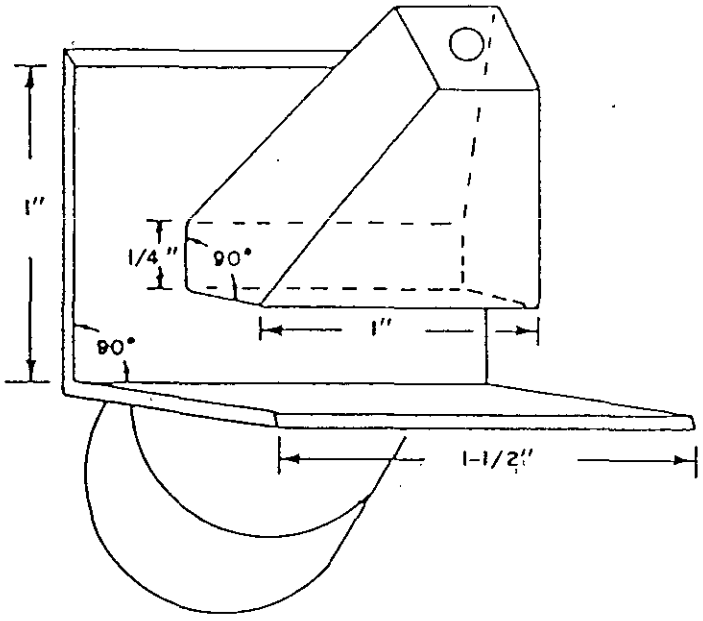


FIGURE 1. *Dielectric test electrodes 90 degree V-shaped.*

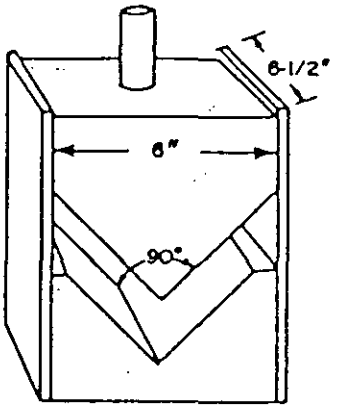


FIGURE 2. *Bending jig, 90 degree V-shaped.*

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4.6.4.4 Method.

4.6.4.4.1 Straight. Specimen size shall be 5 by 5 inches. Specimens shall be conditioned 168/23/55 and 168/23/96. Dielectric breakdown shall then be measured on the specimens at room conditions of 23 °C within 10 minutes after removal from the humidity conditioning chamber. The specimens shall be maintained at 220 °C for 8 ± 3 minutes, and then shall be subjected to dielectric breakdown at this temperature.

4.6.4.4.2 Bent. Specimens having a minimum size of 3 by 3-1/2 inches shall be cut and conditioned 168/23/55. Specimens shall be bent lengthwise and others crosswise using the bending jig (see 4.6.4.3). Each of the specimen shall then be placed in the V-shaped electrodes (see 4.6.4.2.2) and subjected to dielectric breakdown. A force of 6 pounds shall be exerted by the upper electrode on the specimen.

4.6.4.5 Calculation of dielectric strength. The dielectric strength in volts per mil shall be calculated by dividing the average dielectric breakdown by the average thickness as determined in 4.6.1.

4.6.5 Volume resistivity.

4.6.5.1 Apparatus. The apparatus shall be in accordance with ASTM D 257 with a dc potential of 100 to 200 volts applied to the specimen.

4.6.5.2 Test electrodes. Test electrodes shall be conductive rubber electrodes used under pressure with a guarded circuit. The electrodes shall be 1 inch in diameter on both sides. A pressure of 25 lb/in^2 shall be applied to the electrode to ensure adequate contact of the electrodes with the specimen.

4.6.5.3 Test method. The electrodes shall be applied to specimens, 3-1/4 by 3-1/4 inches, and shall be centered with respect to each other. The volume resistance between the electrodes shall be measured at 25 °C after voltage has been applied for 1 minute on each specimen after conditioning 168/23/96.

4.6.5.4 Calculation. Volume resistivity (r) in ohm-centimeters, shall be calculated as follows:

$$r = \frac{RA \times 10^6}{t}$$

Where: R = volume resistance in megohms
A = area of the guarded electrode in cm^2
t = average thickness of specimen in centimeters.

4.6.6 Dissipation factor and dielectric constant.

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4.6.6.1 Test equipment. Bridge, generator, and detector shall be as specified in ASTM D 150 for low frequency measurements.

4.6.6.2 Test specimens. Test specimens, 3-1/4 by 3-1/4 inches, shall be used with 1-inch diameter pressure electrodes and a guarded circuit as specified in 4.6.5.2.

4.6.6.3 Test specimen method. Specimens shall be conditioned 168/23/96. One specimen at a time shall be removed from the conditioning chamber, and measurements shall immediately be made at 23 °C at a frequency of 60 Hz. Other similar specimens shall be heated to 220 °C for 15 ± 5 minutes and determinations shall be made while at this temperature. Individual dissipation factor and dielectric constant values shall be calculated for each of the specimens. These values shall be averaged for each condition. Calculations and edge corrections shall be made in accordance with ASTM D 150.

4.6.7 Tensile strength.

4.6.7.1 Apparatus. Tensile strength shall be determined on a motor-driven, constant rate or pendulum-type of jaws separation testing machine.

4.6.7.2 Specimens. The test specimens shall be cut to conform to ASTM D 828. Specimens cut lengthwise and cut crosswise from the sheet shall be measured at 23 °C after conditioning 168/23/55.

4.6.7.3 Method. Each test specimen shall be placed in the testing machine and stretched until it breaks. The rate of separation of the jaws of the machine shall be not greater than 12 ± 1/2 inches per minute. The average tensile strength shall be in pounds per inch width.

4.6.8 Thermal endurance. The thermal endurance evaluation shall consist of tests on the 10 mil thick material only, using the method specified in ASTM D 1830.

4.6.9 Elevated temperature exposure.

4.6.9.1 Test specimens. Five specimens shall be prepared as specified in 4.6.9.2 for each thickness of material.

4.6.9.2 Method. The specimens shall be aged for 10 days at 300 °C for types I and II and at 240 °C for types III and IV. The specimens shall then be removed from the oven, cooled, and tested for dielectric breakdown in accordance with 4.6.4. The dielectric strength in volts per mil shall be calculated by dividing the average dielectric breakdown by the average thickness as determined in 4.6.1.

4.6.10 Bursting strength.

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4.6.10.1 Apparatus. Bursting strength shall be determined on a motor-driven mullen burst tester.

4.6.10.2 Specimens. The test specimens shall be cut to conform to ASTM D 774. Specimens shall be measured at 23 °C after conditioning 168/23/55.

4.6.10.3 Method. Each test specimen shall be placed on the mullen burst tester and stretched to the bursting point. The diaphragm shall be retracted. The average mullen burst of both sides of the paper shall be in lb/in^2.

4.6.11 Loss on ignition.

4.6.11.1 Apparatus. Loss on ignition shall be tested in a muffle furnace constructed to operate at 925 ± 25 °C.

4.6.11.2 Specimen. The test specimen shall be prepared in accordance with TAPPI T413 OM-85.

4.6.11.3 Method. The specimen weight shall be determined after drying. The specimen shall be placed in a muffle furnace at 925 ± 25 °C for 1 hour. The weight loss shall then be calculated after cooling. Percent loss on ignition shall be calculated.

4.6.12 Apparent thermal conductivity.

4.6.12.1 Apparatus. Thermal conductivity shall be tested on an apparatus as specified in ASTM C 177.

4.6.12.2 Specimen. The specimen shall be 10 mils in thickness. The shape of the specimen shall be a 200 mm disk. Several disks may be required for completing this test.

4.6.12.3 Method. The specimen shall be evaluated in accordance with ASTM C 177. The apparent thermal conductivity shall be computed from:

$$K = (q\text{-}A)\ \frac{(\Delta X)}{\Delta T}$$

Where: ΔX = Thickness of specimen
ΔT = Temperature difference across specimen
q = Power dissipation in main heater
A = Meter surface area
K = Apparent thermal conductivity

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4.7 Inspection of packaging. Sample packages and packs, and the inspection of the preservation, packing, and marking for shipment, stowage, and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition.)

5.1 General.

5.1.1 Navy fire-retardant requirements.

a. *Treated lumber and plywood.* Unless otherwise specified (see 6.2), all lumber and plywood including laminated veneer material used in shipping container and pallet construction, members, blocking, bracing, and reinforcing shall be fire-retardant treated material conforming to MIL-L-19140 as follows:

Levels A and B	–	Type II – weather resistant Category 1 – general use
Level C	–	Type I – non-weather resistant Category 1 – general use.

b. *Fiberboard.* Unless otherwise specified (see 6.2), fiberboard used in the construction of class-domestic, non-weather resistant fiberboard and cleated fiberboard boxes including interior packing forms shall meet the flamespread index and the specific optic density requirements of PPP-F-320, and amendment thereto.

5.2 Preservation. Preservation shall be level A, C, or commercial as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Form S. Sheets (see 3.3.2.2 and 6.5) shall be unit protected in accordance with method IC of MIL-P-116 as follows. The sheets shall be stacked in like thickness and size in quantities specified (see 6.2) with a sheet of fiberboard the same size as the sheets placed above and under the stack and secured in place with tape conforming to PPP-T-60. The stacks shall be individually wrapped with waterproof barrier material conforming to PPP-B-1055 with all folds, seams, and laps sealed with water resistant adhesive conforming to MMM-A-260.

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5.2.1.2 Forms R and T. Rolls (see 3.3.2.1 and 6.4) shall be unit protected in accordance with sub-method IC–3 of MIL-P-116 and individually unit packed in a water resistant folding, set-up, or metal edged paperboard box meeting the requirements of MIL-STD-2073-1, table I, selection and closure of unit and intermediate containers, appendix F, at the contractors option (see 6.2). Box closure shall be in accordance with the applicable box specification.

5.2.1.2.1 Intermediate pack. Rolls of like description and size, shall be intermediate packed in snug-fitting, class-weather resistant, fiberboard boxes meeting the requirements of MIL-STD-2073-1, table I, selection and closure of unit and intermediate containers, appendix F.

5.2.2 Level C. Material in the form specified shall be preserved as specified under level A except that the unit containers specified in 5.2.1.2 and 5.2.1.2.1 shall be as follows:

a. The paperboard containers shall be of the domestic or non-weather resistant type, class or variety as applicable and,

b. The fiberboard box shall be of the class-domestic/fire-retardant material (see 5.1.1b). The box enclosure shall be in accordance with method I using pressure sensitive adhesive tape.

5.2.3 Commercial. Commercial preservation, and unit pack shall be in accordance with ASTM D 3951.

5.3 Packing. Packing shall be level A, B, C, or commercial as specified (see 6.2).

5.3.1 General requirements for levels A, B, and C. Containers selected (see 5.3.2), shall be of minimum weight and cube consistent with the protection required, and of uniform size.

5.3.2 Levels A, B, and C containers. Materials preserved as specified (see 5.2), shall be packed in exterior shipping containers for the level of packing specified (see 5.3), in accordance with table VII of MIL-STD-2073-1, appendix C, and herein. Unless otherwise specified (see 6.2), container selection shall be at the contractor's option.

5.3.2.1 Caseliners, closure, and gross weight.

5.3.2.1.1 Caseliners. Unless otherwise specified (see 6.2), level A shipping containers containing items preserved level C or commercial shall be provided with waterproof caseliners in accordance with MIL-STD-2073-1.

5.3.2.1.2 Closure. Container closure, reinforcing, or banding shall be in accordance with the applicable container specification or appendix thereto except that weather-resistant fiberboard boxes shall be closed in accordance with method V and reinforced with non-metallic or tape banding and domestic or fire-retardant fiberboard boxes shall be closed in accordance with method I using pressure sensitive tape.

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5.3.2.1.3 Weight. Wood, plywood, and cleated type containers exceeding 150 pounds gross weight shall be modified by the addition of skids in accordance with MIL-STD-2073-1 and the applicable container specification or appendix thereto.

5.3.3 Commercial. Material preserved as specified (see 5.2) shall be packed for shipment in accordance with ASTM D 3951 and herein.

5.3.3.1 Container modification. Shipping containers exceeding 200 pounds gross weight shall have a minimum of two, 3-inch by 4-inch nominal wood skids laid flat, or a skid or sill type base which will support the material and facilitate handling by mechanical handling equipment during shipment, stowage, and storage.

5.4 Marking, levels A, B, C, and commercial. In addition to any special marking required (see 6.2), and herein, interior (unit and intermediate) packs and shipping containers shall be marked including bar coding for shipment, stowage, and storage in accordance with MIL-STD-2073-1, appendix F. In addition, interior packs and exterior shipping containers shall be marked "INSULATION PAPER ASBESTOS FREE (see 3.2)".

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The material covered by this specification is intended for use in generator, motor and transformer coils, slot liners, ground insulation, phase-to-phase insulation, phase and end insulation, primary-secondary separation, coil exteriors, coil wrapper tapes, and similar applications. The highest temperature (limiting "hottest spot" temperature) to which the insulation may be subjected continuously with normal life expectancy is 220 °C for types I and II and 180 °C for types III and IV. In general, operating voltage stress should be below 50 volts per mil of thickness.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification

b. Type and form required (see 1.2)

c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1 and 2.2)

d. When first article is required (see 3.1)

e. Nominal thickness (see 3.3.1)

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f. Width and length of rolls of forms R and T material (see 3.3.2.1 and 6.4)

g. Whether or not form S material is to be provided trimmed as either flat sheets or rolls (see 3.3.2.2).

h. Width and length of form S material (see 3.3.2.2)

i. Level of preservation and level of packing required (see 5.2 and 5.3)

j. Quantity square yards (1 square yard = 36 by 36 inches) (see 6.5)

k. Special marking is required (see 5.4)

l. Container selection if other than contractors option (see 5.2.1.2 and 5.3.2)

m. When caseliners are not required (see 5.3.2.1.1).

6.3 First article. When first article inspection is required, the contractor officer should provide specific guidance to offerors whether the items(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first production items, or a standard production item from the contractor's current inventory (see 3.1), and the number of items to be tested as specified in 4.3. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or tests, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Rolls of standard width and put-up. Standard width and put-up is as follows:

a. Widths: 12-, 24-, and 36-inch

b. Standard put-up is on 3-inch id core with 11-1/2 ± 1/2-inch od for both form R and form T rolls; 6-inch id core for 0.040, 0.050, and 0.060

c. The approximate linear yards contained on rolls of standard put-up is shown in table VI

d. The following thicknesses are usually supplied as sheets: 0.040, 0.050, and 0.060 type II.

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TABLE VI. *Lengths on standard rolls.*

Nominal insulation thickness (inches)	Approximate linear yards 3-inch core and 11-1/2 inch od roll
0.0025	960
0.003	800
0.0033	720
0.0045	530
0.0075	370
0.005	480
0.007	350
0.010	260
0.015	170
0.020	130
0.025	105
0.030	90

6.5 Sheets of standard sizes and put-up, form S. Standard width and put-up is as follows:

a. Sizes: 36 by 36 and 24 by 36 inches
b. Sheets per carton as shown in table VII.

TABLE VII. *Sheets per carton.*

Nominal thickness (inches)	Sheets per carton	
	36 by 36	*24 by 36*
0.0025, 3, 3.3	1000	1000
0.005, 0.0045	500	550
0.007, 0.0075	300	500
0.010	200	300
0.015	150	200
0.020	100	150
0.025	100	100
0.030	50	100
0.040	50	75
0.050	50	75
0.060	50	75

6.6 Approximate areas per pound. The approximate areas per pound of the unfired ceramic paper covered by this specification are shown in table VIII.

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TABLE VIII. *Areas per pound, type I and II only (approximate).*

Nominal insulation thickness (inches)	Area in square yards per pound
0.005	4.5
0.007	2.9
0.010	2.1
0.015	1.4
0.020	1.1
0.025	0.9
0.030	0.6
0.040	0.5
0.050	0.4
0.060	0.3

6.7 Part or identifying number (PIN). Material should be identified and ordered by the standard system, M24683-IR05S100. The part numbering system shall be as follows:

M24683-1R05A525

Material quantity per roll or package:
Use three digits – include leading zeros
For forms R and T = average yards per roll
For form S = sheets per package

Material size – as specified in table I (use one letter)

Material thickness: Use two digits – include leading zero
Digits indicate thickness in thousandth of an inch

Material form – (use one letter):
R = form R, S = form S, T = form T

Material type – (use one digit): 1 = type I, 2 = type II
3 = type III, 4 = type IV

Specification number – (use five digits)

Specification identified (use one letter):
B = military specification for materials in bulk quantities
D = military specification using metric measurement units
M = military specification using US customary units.

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6.7.1 Part numbering system size. The part numbering system size shall be as specified in table IX.

6.8 Subject term (key word) listing.

Aluminum silicate
Asbestos
Ceramic
Form R
Form S
Form T

TABLE IX. *Part numbering system size.*

Size code letter	Form S sheet size (inches)	Form R roll width (inches)	Form T roll width (inches)
A	24 x 36	4	3/8
B	36 x 36	6	1/2
C		8	5/8
D		10	3/4
E		12	7/8
F		14	1
G		16	1-1/8
H		18	1-1/4
J		20	1-3/8
K		22	1-1/2
L		24	1-5/8
M		26	1-3/4
N		28	1-7/8
P		30	2
Q		32	2-1/4
R		34	2-1/2
S		36	2-3/4
T		38	3
U		40	3-1/4
V		42	3-1/2
W		44	3-3/4
X		46	3-15/16
Y		48	
Z	Special[1]	Special[1]	Special[1]

[1]Special Z size code requires a separate remarks entry to provide desired size information. Using size code Z forms a non-standard part number that cannot be handled by most automated systems and therefore should be avoided when possible.

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6.9 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Preparing activity:
Navy – SH
(Project 5970-N670)

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	MIL-I-24638A(SH)	1990 November 26

3. DOCUMENT TITLE

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
MR. C.Y.LU (CODE 56Z32)	(1) Commercial (703) 602-3124	(2) AUTOVON 332-3124
c. ADDRESS (Include Zip Code) COMMANDER, NAVAL SEA SYSTEMS COMMAND DEPARTMENT OF THE NAVY WASHINGTO, DC 20362-5101	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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